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Detectors and Accelerators in Particle Physics
— Final report —

Geant4 Simulation of the Bragg Peak for 100 MeV Protons in Water

A comparison of physics lists

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Academic Year 2025–2026

1 Introduction

Protons are widely used in radiotherapy because of the characteristic shape of their depth–dose profile: the energy deposited per unit length rises sharply near the end of the particle range, producing the so-called *Bragg peak*, while the dose released beyond it drops almost to zero, allowing a highly localised energy release in the target volume while sparing the surrounding tissue. In this work we use the Monte Carlo toolkit GEANT4 [1] to simulate a monoenergetic beam of 100 MeV protons impinging on a block of water, we reconstruct the Bragg curve, and we compare the results obtained with several of the available *physics lists*.

1.1 Proton energy loss and the Bragg peak

A proton crossing matter loses energy mainly through inelastic Coulomb collisions with the atomic electrons of the medium, at the average rate given by the Bethe–Bloch formula,

$$-\frac{dE}{dx} = Kz^2 \frac{Z}{A} \frac{1}{\beta^2} \left[\frac{1}{2} \ln \frac{2m_e c^2 \beta^2 \gamma^2 T_{\max}}{I^2} - \beta^2 - \frac{\delta(\beta\gamma)}{2} \right], \quad (1)$$

where z is the projectile charge, Z/A the atomic number to mass number ratio of the medium, I its mean excitation energy, $T_{\max}$ the maximum energy transfer in a single collision, δ the density effect correction and $K \approx 0.307 \text{ MeV cm}^2/\text{mol}$. Because of the $1/\beta^2$ term, the stopping power increases steeply as the proton slows down, so most of the energy is released just before the particle comes to rest: this produces the Bragg peak. The depth at which the proton stops defines the *range*; the continuous-slowing-down-approximation (CSDA) range in water at 100 MeV is approximately 7.7 cm [4, 5], which provides the benchmark for validating the simulation. Two further effects shape the realistic profile: *range straggling*, the statistical spread of the individual stopping points that smears the otherwise sharp peak, and *nuclear interactions*, which remove primary protons from the beam, reduce the peak-to-plateau ratio and generate secondary particles (protons, neutrons, γ rays, fragments). The latter are precisely where different hadronic physics lists are expected to differ.

1.2 Objectives of this work

The goals of the simulation are:

1. to reconstruct the depth–dose distribution (Bragg curve) of 100 MeV protons in water;
2. to extract characteristic quantities such as the peak position, the practical range and the distal fall-off, and to compare them with reference tabulated values [4];
3. to repeat the simulation with different GEANT4 physics lists and assess their impact on the peak position, peak height and on the secondary-particle contribution, identifying the most suitable choice for this process.

2 Simulation setup

The simulation was developed starting from the GEANT4 example AnaEx01 by C. Mancini Terracciano [3], replacing the original multi-layer calorimeter with a single homogeneous block of water. The application was run with GEANT4 version 11.4 (patch 01) and the standard physics-dataset files distributed with the toolkit.

2.1 Geometry, beam and scoring

The detector is a single rectangular box of water (G4_WATER, density 1.000 g/cm³) inside a slightly larger world volume of air (G4_AIR); the relevant parameters are summarised in Table 1. The primaries are 100 MeV protons, generated by a G4ParticleGun as a mono-directional pencil beam entering through the centre of the front face along + z ; 5×10^4 protons were simulated for each configuration, with the same seed for every physics list.

Parameter	Value
Material	water (G4_WATER)
World material	air (G4_AIR)
Transverse size ($x \times y$)	6×6 cm ²
Longitudinal size (z)	10 cm
Scoring bin width (raw)	0.25 mm
Analysis bin width	0.5 mm
Maximum step length in water	0.1 mm

Table 1: Geometrical and scoring parameters of the water phantom.

The depth–dose profile is reconstructed in the user stepping action by assigning the energy deposited in every step to a longitudinal bin according to the z of the step mid-point (transverse coordinates integrated over); the binning is a purely software scoring grid. A step-length limit of 0.1 mm (G4UserLimits with G4StepLimiterPhysics) keeps the tracking steps well below the bin width, and the raw 0.25 mm histogram is re-binned to 0.5 mm to suppress a residual aliasing pattern from the finite step size. After each run the histogram is normalised to the number of primaries and written to CSV; the characteristic quantities of Sec. 3 are obtained from the re-binned profile by local interpolation.

To quantify the statistical uncertainties, the reference QGSP_BIC configuration was run five times with independent random seeds: the run-to-run scatter is 0.009 mm on the practical range R_{80} , 0.07 MeV on the leaked energy E_{leak} and 0.02 on the peak-to-entrance ratio, while the peak position does not move at the 0.5 mm analysis resolution. These figures are used in Sec. 3 to judge which differences between physics lists are significant.

2.2 Physics lists

A *physics list* in GEANT4 combines an **electromagnetic** component, responsible for the ionisation energy loss and hence for the position and shape of the Bragg peak, and a **hadronic** component, which governs the nuclear interactions and the production of secondaries. Six reference lists sharing the same standard electromagnetic option were compared, spanning the three cascade families relevant at this energy:

- QGSP_BIC – Binary Cascade for nucleon inelastic interactions below a few hundred MeV; the list most often recommended for proton therapy, adopted here as the reference.
- QGSP_BIC_HP – as above, with high-precision (HP) neutron models tracking thermal and epithermal neutrons.
- QGSP_BERT and FTFP_BERT – Bertini intra-nuclear cascade at low energy; identical results at 100 MeV (Sec. 3.2).
- QBBC – general-purpose list combining Binary and Bertini elements, tuned also for medical physics.

- FTFP_INCLXX – Liège cascade (INCL++), an independent alternative for proton interactions.

To probe the sensitivity to the electromagnetic modelling as well, a seventh configuration, denoted QGSP_BIC_EMZ, combines the QGSP_BIC hadronic physics with the more accurate G4EmStandardPhysics_option4 constructor: comparing it with the reference isolates the effect of the electromagnetic option on the reconstructed range, at fixed hadronic modelling.

3 Results and discussion

3.1 Reconstruction of the Bragg curve

Figure 1 shows the depth–dose distribution obtained with the reference QGSP_BIC physics list. The curve reproduces the expected features: a slowly rising *plateau* in the entrance region, a pronounced *Bragg peak* near the end of the range, and a steep *distal fall-off* beyond which the deposited energy drops almost to zero.

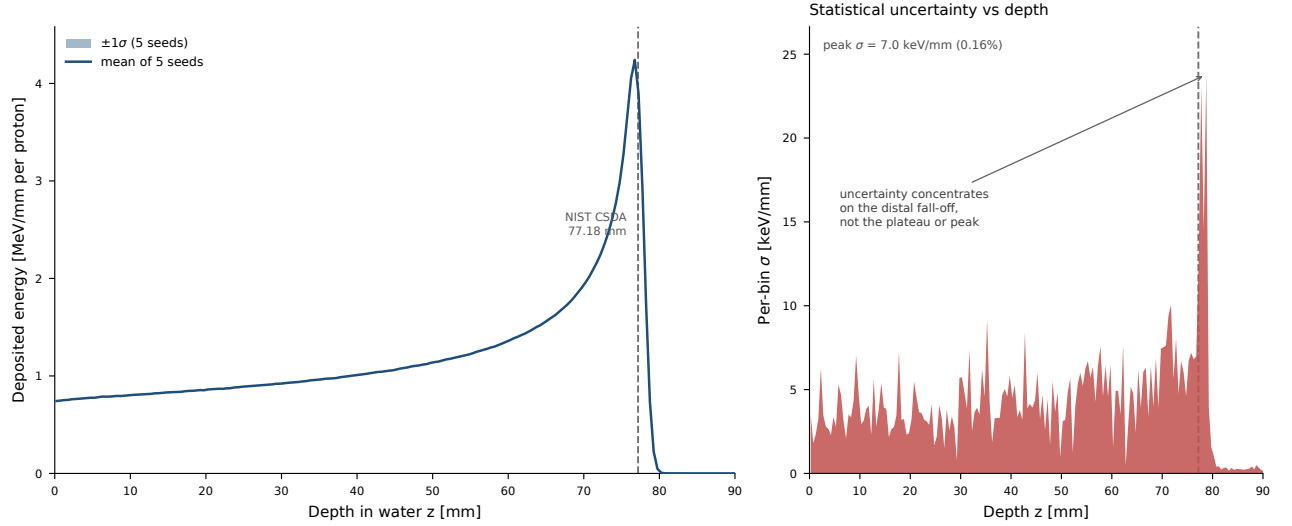


Figure 1: *Left*: depth–dose distribution (Bragg curve) for 100 MeV protons in water (QGSP_BIC); the line is the mean of five independent runs of 5×10^4 primaries and the $\pm 1\sigma$ band is thinner than the line width on this scale. The dashed line marks the CSDA range from PSTAR [4]. *Right*: per-bin statistical uncertainty $\sigma(z)$, at the 0.16 % level across plateau and peak and concentrated on the steep distal fall-off.

From the simulated curve we extract the quantities in Table 2. Following the convention of clinical proton dosimetry, the practical range is identified with the distal 80% point R_{80} , which for a mono-energetic beam is essentially insensitive to range straggling and is therefore a robust estimator of the mean projected range. The peak is located at $z_{\text{peak}} = 76.75$ mm and the practical range at $R_{80} = 77.53$ mm, about 0.4% above the tabulated CSDA value of 77.18 mm [4, 5] – an agreement consistent with the difference in definition between the two quantities, which supports the validity of the electromagnetic modelling of the energy loss. An independent check is provided by the entrance region: the mean energy deposited over the first few millimetres, 0.759 MeV/mm, agrees to about 4% with the PSTAR stopping power of 0.729 MeV/mm at 100 MeV [4].

Quantity	Value
Bragg peak position z_{peak}	76.75 mm
Distal 90% range R_{90}	77.31 mm
Practical range R_{80} (distal 80%)	77.53 mm
Distal 50% range R_{50}	78.08 mm
Distal fall-off (80% $\rightarrow$ 20%)	1.16 mm
Full width at half maximum	6.64 mm
Peak to entrance dose ratio	5.6
Total energy deposited per proton	97.7 MeV
Reference CSDA range [4, 5]	77.18 mm

Table 2: Characteristic quantities extracted from the simulated Bragg curve for the reference QGSP_BIC physics list.

3.2 Comparison of physics lists

The simulation was repeated with the physics lists of Sec. 2.2, keeping all other parameters fixed. Figure 2 overlays the resulting Bragg curves, with a magnified view of the peak region, and Table 3 summarises the figures of merit. The curves for QGSP_BERT and FTFP_BERT were identical to within the numerical precision of the output and are represented by a single line. The QGSP_BIC_EMZ configuration, which differs from the reference only in the electromagnetic option, gives a practical range within 0.02 mm of QGSP_BIC – below the run-to-run scatter of Sec. 2.1 – so the more elaborate electromagnetic models leave the reconstructed range essentially unchanged at this energy.

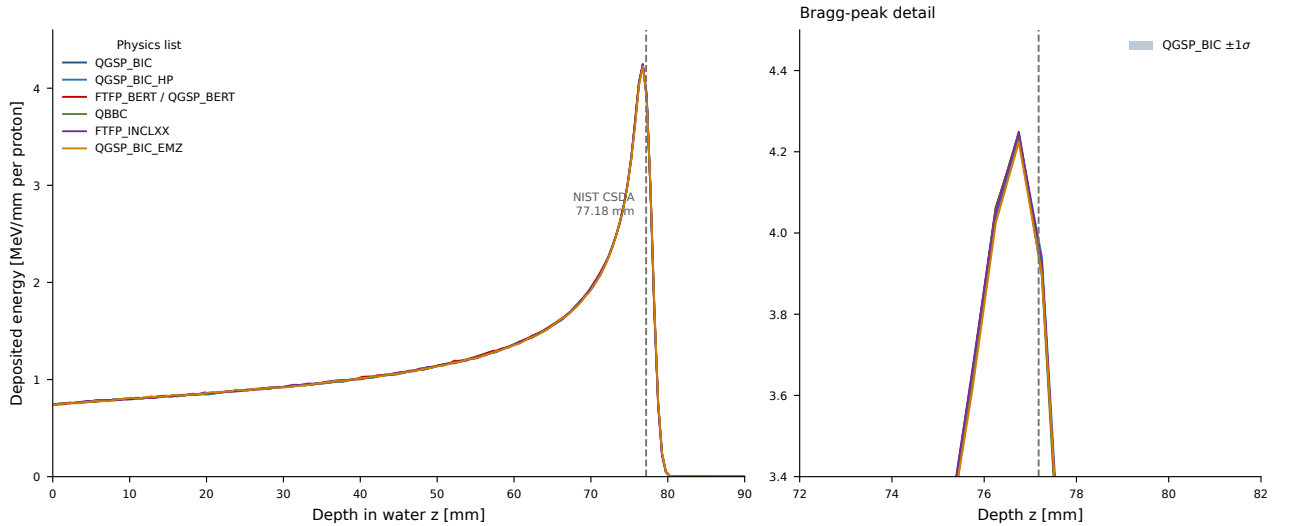


Figure 2: Depth-dose distributions obtained with the different physics lists. Left: full Bragg curves, visually indistinguishable on this scale. Right: magnified view of the peak region. The dashed line marks the CSDA range [4].

Physics list	z_{peak} [mm]	R_{80} [mm]	Peak/entr.	E_{leak} [MeV]
QGSP_BIC	76.75	77.53	5.58	2.29
QGSP_BIC_HP	76.75	77.54	5.58	2.30
QGSP_BERT	76.75	77.52	5.62	2.04
FTFP_BERT	76.75	77.52	5.62	2.04
QBBC	76.75	77.51	5.59	2.28
FTFP_INCLXX	76.75	77.52	5.59	2.12
QGSP_BIC_EMZ	76.75	77.53	5.58	2.35

Table 3: Figures of merit for the different physics lists: peak position, distal 80% range, peak to entrance dose ratio and energy escaping the phantom per primary proton (E_{leak} , carried mostly by neutrons and γ rays). The last row uses the reference hadronic physics with the `option4` electromagnetic constructor. The run-to-run statistical scatter (five independent seeds, QGSP_BIC) is 0.009 mm on R_{80} , 0.07 MeV on E_{leak} and 0.02 on the peak to entrance ratio.

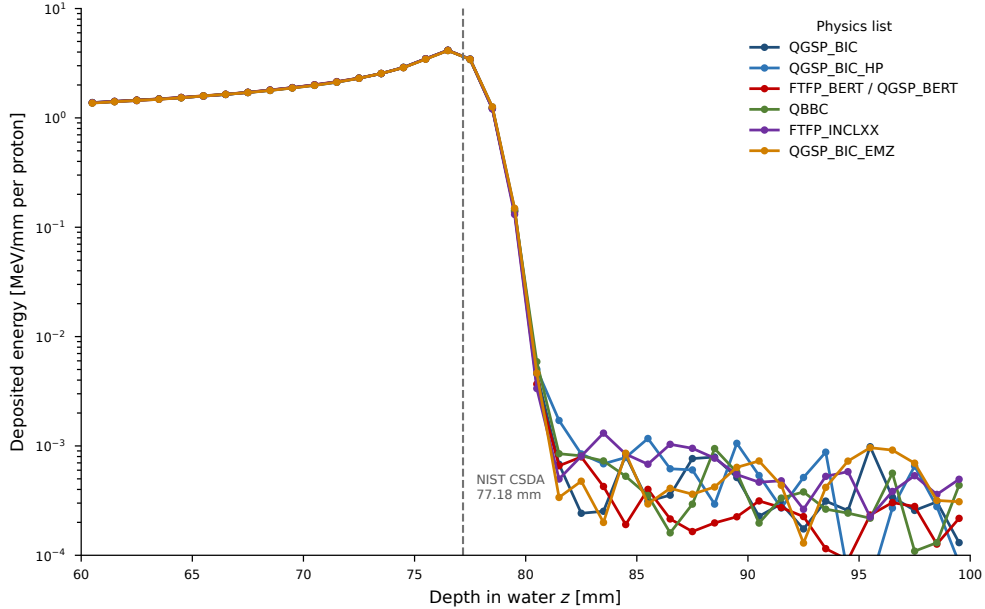


Figure 3: Depth-dose distributions in the peak and post-peak region on a logarithmic scale, re-binned to 1 mm. The distal fall-off is common to all physics lists; the lists diverge only in the secondary-particle tail beyond about 81 mm, three to four orders of magnitude below the peak, where the present statistics become limiting.

3.3 Discussion

The peak position and the plateau are essentially identical across all physics lists: the spread of z_{peak} is below the 0.5 mm analysis bin and that of R_{80} is only about 0.03 mm. This is expected, since the depth at which the beam stops is fixed by the electromagnetic energy loss, common to all lists. Only the *position* of the peak, however, is a purely electromagnetic quantity: the peak *height* also depends on the nuclear interactions, since inelastic reactions remove primary protons before they reach the end of range, so a list predicting more nuclear absorption deposits slightly less energy at the peak and more in the secondary-particle tail.

The differences between the lists indeed appear in the quantities controlled by the hadronic component. A convenient global indicator is the energy escaping the phantom per primary proton, E_{leak} , carried mainly by neutrons and γ rays from non-elastic nuclear reactions. As shown in Table 3, the Binary-Cascade lists and QBBC predict the largest leakage (2.28 MeV to 2.30 MeV per proton), the Bertini-based lists the smallest (2.04 MeV) and FTFP_INCLXX an intermediate value (2.12 MeV). Against the run-to-run scatter of 0.07 MeV, this gap of about 0.25 MeV ($\sim 10\%$) corresponds to several standard deviations and is a genuine systematic difference between the models; the variation of the peak to entrance ratio (5.58 to 5.62), only about twice its statistical scatter, is at most marginally significant. The electromagnetic option has no discernible effect on these hadronic indicators. The residual imprint of the hadronic model on the profile itself is confined to the distal tail (Fig. 3): the fall-off is common to all lists, and only beyond about 81 mm, at the 10^{-3} level relative to the peak, do the lists begin to differ, in line with their different secondary yields.

In summary, the range and plateau of the Bragg curve of 100 MeV protons are largely insensitive to the physics list, whereas the peak height, the distal tail and the secondary-particle production are not. For applications in which the latter matters, QGSP_BIC appears to be the most suitable choice: the Binary Cascade has been validated for protons below a few hundred MeV and is the model most commonly recommended for hadron-therapy studies [6]. Its QGSP_BIC_HP variant is preferable when thermal-neutron transport or activation is of interest, while FTFP_INCLXX provides a reasonable independent alternative.

4 Conclusions

We have simulated the interaction of a 100 MeV proton beam with a water phantom using GEANT4 and reconstructed the Bragg curve. The peak is located at 76.75 mm and the practical range at $R_{80} = 77.53$ mm, in agreement to within about 0.4% with the tabulated CSDA value of 77.18 mm [4, 5]; the entrance dose likewise agrees with the PSTAR stopping power, confirming that the electromagnetic energy-loss processes, which determine the range, are adequately modelled.

Comparing six hadronic physics lists, together with a variant using the more accurate option4 electromagnetic constructor, we found that the range is essentially independent of both the hadronic model and the electromagnetic option: R_{80} agrees to within 0.03 mm, comparable with the 0.009 mm run-to-run scatter estimated from five independent seeds. The hadronic list mainly affects the energy leaving the phantom as secondary neutrons and γ rays, which varies by about 10% between the models – several times the statistical scatter, and thus a genuine systematic effect – being largest for the Binary-Cascade lists and smallest for the Bertini-based ones. Among the lists tested, QGSP_BIC offers a suitable compromise between physical accuracy at this energy and computational cost, in line with the standard recommendations for hadron therapy; QGSP_BIC_HP is advisable when detailed neutron transport is required.

Possible extensions of this work include studying the dependence of the range on the beam energy, building a spread-out Bragg peak (SOBP) by superimposing beams of different energies, investigating the lateral dose profile, and repeating the analysis in different materials.

References

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